Synthetic-flux control of hyperchaos order in a dissipative optomechanical model

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Abstract

A gauge-invariant synthetic-flux phase controls the *order* of a drive- and coupling-gated hyperchaos transition in a six-dimensional dissipative model. The full six-exponent Lyapunov spectrum resolves up to four simultaneously positive directions, beyond reported single-mode quadratic-optomechanics benchmarks. The route is drive-gated: a Neimark–Sacker onset at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$), re-stabilization at the sampled intermediate drive ($E = 2.0$, count near zero), and finite-time positive-exponent counts reaching three at $E = 4.0$ and four at $E = 8.0$. A phase-consistent Floquet–Lyapunov protocol (monodromy multipliers, dissipative volume balance $\sum \lambda_i = \mathrm{Tr}(J) = -1.04$, and a 432-run classification ensemble spanning twelve seeds, twelve phases, and three drives) supports the classification under the stated protocol, and orbit continuation localizes the onset. As a secondary model-level clarification, the matched-resource force-sensing protocol returns a null gain on the chaotic attractor (flux-on Fisher information 0.54× flux-off at the resolved $E = 4$ points) and only a modest order-unity gain (1.039 ± 0.014) on the stable orbit, consistent with the matched-Fisher lemma. Truncated-Fock and truncated-Wigner checks support the mean-field description at selected points. All results are strictly model-level—the sector lies 2542× beyond anchored silicon couplings, and closing that gap requires ultrasonic characterization of the mechanical degeneracy ω_2, a measured inter-resonator hopping J_m, and a genuine gigahertz platform.

Keywords: hyperchaos, Lyapunov spectrum, Optomechanics, Synthetic gauge flux, Neimark–Sacker bifurcation, Bifurcation control

1. Introduction

The order of a hyperchaos transition—the number of simultaneously positive Lyapunov exponents—quantifies the dimensionality of the unstable manifold and therefore the number of independent directions along which small perturbations grow. In single-mode quadratic optomechanical platforms, this order has so far been limited to two [1]. Whether a synthetic-flux phase can raise that order in a multi-mode radiation-pressure model, and whether the resulting geometric control can be corroborated by full-spectrum methods, had not been resolved. Selecting a desired dynamical regime by tuning a control parameter is a classic theme in nonlinear dynamics [2, 3], with the control of hyperchaos studied across laser systems, electronic circuits, and coupled oscillators [4, 5].

We show, within one normalized six-dimensional model (building on the topology of Muthukumar *et al.* [6]), that the synthetic-flux phase Φ_{syn} reproducibly controls the *order* of a drive- and coupling-gated hyperchaos route—up to four simultaneously unstable directions, beyond reported single-mode optomechanical benchmarks. The onset is localized to a Neimark–Sacker bifurcation and is corroborated by a phase-consistent Floquet–Lyapunov protocol. The route itself is structured rather than monotonic: after the torus onset the orbit nearly re-stabilizes in a window at intermediate drive, and the positive-exponent count then climbs through three up to four as the drive increases. As a secondary geometric clarification, the same matched measurement on the chaotic attractor yields no flux-induced sensing enhancement. These are strictly model-level results: the strong-coupling sector explored here lies far beyond present experimental anchors.

A nonlinear optomechanical platform provides the physical substrate: radiation-pressure nonlinearity, coherent mechanical motion, dissipation, periodic modulation, and a complex hopping amplitude that supports a synthetic gauge structure [7–11], together with Floquet engineering [12–17]. Chaos in optomechanical models is established in single-mode and hybrid configurations [18–20], and hyperchaos has been generated in other driven nonlinear platforms such as delayed-feedback semiconductor lasers [5].

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The central result is shown in Figure 3: the synthetic-flux phase $\Phi_{\rm syn}$ selects the number of simultaneously positive Lyapunov directions in the strongly damped, strongly coupled sector; the count reaches four at $E = 8$, $\theta = -\pi/2$ (seed-dependent; see below), and drops to two at $\theta = 0 \pmod{\pi}$. Relative to prior work, Muthukumar *et al.* [6] established the one-cavity/two-resonator topology and reported phase-selected chaotic regimes at resolved-sideband couplings [21, 22], but did not resolve the *full Lyapunov spectrum* (so that the instability order cannot be counted) nor quantify a flux-dependent advantage under matched resources. The closest full-spectrum report [1] is restricted to quadratic single-mode coupling where no synthetic-flux phase is available. We extend the Muthukumar topology by making the convention $\Delta = -\Delta_{\rm M}$ explicit, allowing $g_1 \neq g_2$, and scanning the strongly damped, strongly coupled sector where the full spectrum resolves up to four simultaneously unstable directions. A second full map at reduced coupling ($g_1/\omega_m = 0.2$) shows the same flux modulation with a lowered ceiling (see Results). We further replace the qualitative sensing proposal by a matched-resource measurement-Fisher protocol [23–27], applied both on the weakly coupled reference orbit and on the chaotic attractor.

The central question is whether the gauge-invariant synthetic-flux phase $\Phi_{\rm syn}$ can act as a reproducible open-loop control coordinate for the *order* of a drive- and coupling-gated hyperchaos transition in a driven dissipative optomechanical model. The answer documented below is yes at the model level: the full six-exponent spectrum resolves up to four simultaneously positive Lyapunov directions, beyond the order reported for quadratic single-mode optomechanics, and the route is localized by a Neimark–Sacker torus bifurcation of the continued drive-locked orbit. Three operational sub-questions structure the analysis. First, does $\Phi_{\rm syn}$ control the hyperchaos order, with the route localized by a continued-orbit scan to a Neimark–Sacker bifurcation? Second, do monodromy multipliers and the long-time Lyapunov spectrum agree on the same periodic orbit, thereby validating the classification? Third, as a secondary geometric bound, after thermal, vacuum, and detector noise are included, does a matched-resource measurement statistic retain a flux-dependent signature on the stable orbit or on the chaotic attractor? A positive answer to the first two questions constitutes the primary claim of this work; the null answer to the third supplies a matched-Fisher bound showing that additional unstable directions do not automatically yield metrological gain. Both results are strictly model-level.

2. Model and conventions

The one-cavity, two-mechanical-resonator topology and phase-dependent hopping are due to Muthukumar *et al.* [6]; the rotating-frame Hamiltonian used here is

$$\frac{H}{\hbar} = -\Delta a^\dagger a + \sum_{j=1}^{2} \omega_j b_j^\dagger b_j - \sum_{j=1}^{2} g_j a^\dagger a (b_j + b_j^\dagger) + J\left(e^{i\theta} b_1^\dagger b_2 + e^{-i\theta} b_2^\dagger b_1\right) + i(E a^\dagger - E^* a), \tag{1}$$

with cavity mode a and mechanical modes b_j. We use $\Delta = \omega_c - \omega_L$ (so $\Delta = -\Delta_{\rm M}$ of Ref. [6], which takes the opposite sign), quadratures $[X, Y] = i$, and the drive $E = \sqrt{\kappa}\,\alpha^{\rm in}$ in the shared normalization. The present analysis allows $g_1 \neq g_2$ and scans the strongly damped, strongly coupled sector $\kappa \sim \gamma \sim g \sim 1 \times 10^{-2}$–$1\,\omega_m$ in which the hyperchaotic attractor is resolved. **A side-by-side comparison with Ref. [6] is given in Table S1.**

The semiclassical factorization gives

$$\dot{\alpha} = \left(i\Delta - \frac{\kappa}{2}\right)\alpha + i\sum_j g_j(\beta_j + \beta_j^*)\alpha + E(t), \tag{2}$$

$$\dot{\beta}_1 = -\left(i\omega_1 + \frac{\gamma_1}{2}\right)\beta_1 + i g_1 |\alpha|^2 - iJe^{i\theta}\beta_2, \tag{3}$$

$$\dot{\beta}_2 = -\left(i\omega_2 + \frac{\gamma_2}{2}\right)\beta_2 + i g_2 |\alpha|^2 - iJe^{-i\theta}\beta_1, \tag{4}$$

with the real state $x = (\alpha_r, \alpha_i, \beta_{1r}, \beta_{1i}, \beta_{2r}, \beta_{2i})^T$. The hopping amplitude is written J in the Hamiltonian and, identically, J_m in normalized ratios ($J_m \equiv J$).

For the three-link loop, the phase coordinate used in the analysis is the gauge-invariant phase

$$\Phi_{\rm syn} = \arg(g_1 J g_2^*) \pmod{2\pi}. \tag{5}$$

A redefinition of the mode phases may redistribute link phases but cannot change $\Phi_{\rm syn}$. In the phase convention of Equation (1) (all phase on the inter-resonator hopping, $g_1, g_2 \in \mathbb{R}$) the invariant reduces to the hopping phase, $\Phi_{\rm syn} \equiv \theta \pmod{2\pi}$, so the two notations coincide and are used interchangeably in the results; the invariant form is retained because it is the coordinate that survives such a redefinition.

3. Methods

The numerical protocol is organized around three independent consistency layers that must agree before any claim of hyperchaos order is accepted. (i) A drive-locked periodic orbit is located as a fixed point of the one-period Poincaré map; its linear stability is diagnosed by the monodromy matrix (Floquet multipliers). (ii) The same orbit is integrated with QR reorthonormalization to obtain the full six-exponent Lyapunov spectrum; positivity is accepted only when the spectrum, the multipliers and the analytic dissipative trace (Equation (7)) are mutually consistent. (iii) As a secondary geometric check, a matched-resource classical Fisher information is evaluated on both the stable reference orbit and on sampled trajectories of the chaotic attractor, under identical noise and resource assumptions. All three layers are implemented with the same phase convention for Φ_{syn} (Eq. (5)). Thresholds used to count positive exponents (1×10^{-4} weak, 1×10^{-3} strong) and the precise definition of the matched Fisher protocol are stated once here and used uniformly thereafter. The strong threshold 1×10^{-3} is chosen to lie above the scale of the mechanical damping rates ($\gamma/2 \sim 0.01$) while remaining larger than typical QR residuals on the stable reference grid; sensitivity of the integer count n_+ to displacements of this cut by factors of three to five is quantified in the Supplementary Material.

3.1. Phase-augmented dynamics and implementation

For a periodic drive with phase ϕ we write

$$\dot{x} = f(x, \phi; p), \qquad \dot{\phi} = \Omega, \tag{6}$$

with the augmented Jacobian $J_{\mathrm{aug}} = (\partial_x f, \partial_\phi f; 0, 0)$. The analytic Jacobian $J = \partial f/\partial x$ is compared column by column with central finite differences; the implemented dissipative trace

$$\mathrm{Tr}\, J = -\kappa - \gamma_1 - \gamma_2, \tag{7}$$

is an independent implementation check. One-period integrations use DOP853 [28] with relative and absolute tolerances 1×10^{-9} and 1×10^{-11} (max step $T/200$); drive-locked orbits are located by fixed-point iteration of the one-period map to a residual below 1×10^{-8}, and the monodromy is obtained by simultaneous integration of the variational equations.

3.2. Periodic orbit and Floquet analysis

A drive-locked periodic orbit is obtained from the fixed point of the one-period Poincaré map; around it the fundamental matrix satisfies

$$\dot{\Phi}(t) = A(t)\Phi(t), \ \Phi(0) = \mathbb{I}, \ M(T) = \Phi(T), \tag{8}$$

with Floquet rates $\rho_i = T^{-1} \log|\mu_i|$. The same orbit is integrated with QR/Benettin reorthonormalization [29–32] to obtain the full Lyapunov spectrum; a positive exponent is accepted only after agreement with the multipliers, the divergence given by (7), and the bounded-attractor diagnostics. In the full-spectrum classification an exponent is counted as positive only above the thresholds stated at the start of this section (Table S12) whereas the Kaplan–Yorke sum uses strictly positive exponents; the threshold count and the Kaplan–Yorke sum can differ near zero. The Lyapunov QR uses a fixed-step fourth-order Runge–Kutta (RK4) (1×10^{-2} step, QR every ten steps), discarding a 1×10^5 transient and integrating 1×10^6 steps per trajectory for the production spectra, and 2×10^5 and 2×10^6 steps for the Floquet–Lyapunov consistency validation of Section 4. The theorem bounds are established at 2×10^6 steps while production spectra use 1×10^6 steps, additionally supported by the six-point window-convergence probe (Table S11). Orbit location uses adaptive DOP853 and long QR spectra use fixed-step RK4; the quoted Floquet–QR discrepancies are the cross-branch agreement of these two integrator branches. A dedicated step/window study at three headline points (Table S9) leaves every non-marginal count unchanged across step halving and window doubling; only marginal exponents move across the cut, and the production record reproduces bit-identically.

3.3. Theoretical bad-cavity sensitivity screen

To examine the parameter dependence of the reduced equations without implying a hardware calibration, the optical amplitude is eliminated in the bad-cavity limit, with

$$\Omega_{\mathrm{eff}} = \Delta + 2g_1\,\mathrm{Re}(\beta_1) + 2g_2\,\mathrm{Re}(\beta_2), \tag{9}$$

and the adiabatic intensity

$$n_{\mathrm{ss}}(\phi) = |E(\phi)|^2/[(\kappa/2)^2 + \Omega_{\mathrm{eff}}^2], \tag{10}$$

substituted into the two mechanical equations. The reduced equations are compared with the full six-dimensional model in common normalized regimes (max infinity-norm differences 9.27×10^{-7} at $\kappa = 20$ and 2.08×10^{-8} at $\kappa = 100$). A theoretical sensitivity screen samples 16 candidates at 5 replicas $\times$ 5 root-solver starts each, using the one-period Poincaré residual as the acceptance criterion; the two objectives are the local stability margin and the normalized drive cost. This is a multi-start fixed-point study, not a Fisher, power-mapping or basin-capture analysis.

3.4. Secondary geometric bound: observability and Fisher definitions

For a measured record

$$y(t) = h[x(t), t; p] + \nu_{\mathrm{det}}(t), \tag{11}$$

the Gaussian-record classical Fisher information [33–36] reads

$$F_C(\vartheta) = (\partial_\vartheta m)^T \Sigma^{-1} (\partial_\vartheta m) + \tfrac{1}{2} \mathrm{Tr}\left[\Sigma^{-1}(\partial_\vartheta \Sigma)\Sigma^{-1}(\partial_\vartheta \Sigma)\right], \tag{12}$$

with a reference comparison required to match input power, observation time, bandwidth, baths, detector noise and estimator [23–27, 37, 38]. For force sensing the parameter of interest is a weak external force F on mechanical mode 1; the matched references are the flux-off coupled sensor ($\theta = 0$) and the single-mode linear sensor ($J_m = 0$). F_C is not quantum Fisher information (QFI); neither quantity is inferred from a Lyapunov exponent.

3.5. Consistency and matched-measurement statements

Two statements formalize the protocol and are referenced throughout the results; the proofs and longer lemmas are in the Supplementary Material (Section S5).

Consistency criterion 1 (Lyapunov–Floquet consistency criterion)**.** *Consider the phase-augmented periodic flow (6) and a converged periodic orbit $x_p(t)$ with fundamental matrix $M(T)$ (Equation (8)). Let $\rho_i = T^{-1}\log|\mu_i|$ be the Floquet rates sorted in the same convention as the QR/Benettin spectrum $\{\lambda_k\}_{k=1}^N$ obtained by variational integration with reorthonormalization. If (i) the Poincaré residual on the converged orbit is below 5×10^{-8}, (ii) the QR spectrum is integrated for $\geq 2\times10^6$ steps after a 2×10^5 transient with reorthonormalization every ten steps and fixed step 1×10^{-2}, and (iii) the implementation trace satisfies (7), then the three diagnostics describe the same linearization of the same orbit: for every tested phase the maximum absolute discrepancy $|\rho_i - \lambda_{\pi(i)}|$ is bounded by 3×10^{-4} and the mean divergence coincides with $\mathrm{Tr}\,J$ to within 5×10^{-5}, the orbit is exponentially stable if and only if all transverse multipliers satisfy $|\mu_i|<1$, and in that case $\lambda_{\max}<0$. A positive QR exponent is accepted as a chaotic signature only after conditions (i) and (iii) are verified on the same orbit and the same phase and its finite-time count passes the convergence probe of Section S6; condition (ii) fixes the validated integration length at which the bounds above are established.*

Lemma 1 (Matched-measurement Fisher bound)**.** *For two configurations A and B sharing the same input power, observation time, integration window, detector noise, baths and estimator, the matched-measurement gain $g_{A/B} \equiv F_C^{(A)}(\vartheta)/F_C^{(B)}(\vartheta)$ is bounded by the ratio of their mean-shift and covariance contributions to Equation (12),*

$$g_{A/B} = \frac{m_A'^{\mathsf{T}} \Sigma_A^{-1} m_A'}{m_B'^{\mathsf{T}} \Sigma_B^{-1} m_B'}, \tag{13}$$

in the regime where the variance term is sub-dominant (verified by recomputing $F_C^{(N)}$ for $N \in \{1, 5, 20\}$ drive periods and showing the matched gain is unchanged to four digits on the stable reference orbit). The extension of the bound to the chaotic attractor rests on the window-resolved susceptibility–stretching check (Section S17), not on the sub-dominance premise. The flux-induced contribution is then bounded by the linear-response effect of phase-controlled mode coupling, and cannot receive a multiplicative enhancement from trajectory separation alone: noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\max}t}$ as the deterministic displacement.

The two statements together bound any claim of flux-induced advantage: a hyperchaos-order change is operational only after the consistency criterion is satisfied on the orbit carrying it. The secondary matched-resource analysis is consistent with Lemma 1: additional unstable directions do not automatically translate into a sensing advantage under shared noise, observation time and power.

4. Results

4.1. Model, gauge, and Floquet–Lyapunov consistency

The formal statements are in Section S5; evaluated on the stable drive-locked orbit at the reference point, the consistency metrics read: Poincaré residual 8.49×10^{-9}, mean divergence -1.04, maximum absolute Floquet–QR discrepancy 2.68×10^{-4}, terminal QR-block standard deviation 5.3×10^{-9}, and gauge-transformed loop-flux residual below 3×10^{-16}. The same criteria were evaluated over a 27-trajectory consistency grid (nine synthetic-flux phases and three independent initial conditions per phase, with 2×10^6 integration steps and 2×10^5 discarded transient steps); all 27 trajectories satisfied the Floquet–Lyapunov consistency bounds, with a largest observed QR exponent of $-0.010\,02$, a maximum Floquet–QR discrepancy of 2.85×10^{-4}, a terminal block standard deviation reaching at most 6.10×10^{-9}, and a divergence residual at most 4.75×10^{-5}. These results restrict the conclusion to the tested grid and do not establish a universal absence of chaos. The grid-level support is shown in Figure 1.

4.2. Flux-dependent attractor structure

The full six-exponent spectrum is recorded everywhere rather than only its largest member, while the positive-exponent count is a finite-time, threshold-dependent classification near marginal exponents. At weak coupling over nine flux phases and three replicates, the cavity-dominated exponents sit near $-\kappa/2 \approx -0.5$ and the four mechanical exponents near $-\gamma/2 \approx -0.01$; the positive-exponent count is zero throughout. Extending the classification over $E \in [0.2, 8.0]$ (54 drive–phase cases) leaves the largest exponent negative (between -0.043 and -0.010), the count at zero, and the Kaplan–Yorke dimension at zero. This rules out a flux-driven route to chaos within the tested domain (Figure 2).

Repeating the classification at strong optomechanical coupling ($g_{1,2}/\omega_m = (0.3, 0.27)$, $\gamma_{1,2}/\omega_m = 0.02$, $\Delta = -\omega_m$) over twelve flux phases and five drives, with the hyperchaos order defined by the positive-exponent count above a 1×10^{-3} threshold (Figure 3), shows a drive-gated transition: at $E = 0.2$ every exponent is negative; at $E = 1.0$ a positive exponent appears at some phases; at $E = 2.0$ the marginal exponent hovers near zero, a phase-dependent dip reaching full zero-count only at the sampled $E = 2.0$; the count reaches three at $E = 4.0$ and four at $E = 8.0$, with the largest exponent up to 0.3223 and the Kaplan–Yorke dimension up to 4.802. Sampled flanks around the re-stabilization read 10/12 zero-count phases at $E = 1.5$ (max $n_+ = 2$), 12/12 at $E = 2.0$, 5/12 at $E = 2.5$ (max $n_+ = 3$) and 3/12 at $E = 3.0$ (max $n_+ = 2$): a phase-dependent dip of finite width. At fixed strong drive the flux phase modulates the count: at $E = 8.0$ it varies between two and four as θ sweeps $[-\pi, \pi)$, with seed- and drive-dependent extrema: the largest counts occur near the phase quadratures ($\theta \approx \pm\pi/2$) and the lowest lie near $\theta = 0$ and $\pm\pi$. The twelve-seed check finds the integer ranges agreeing between θ and $-\theta$ for 13 of 15 mirrored phase pairs—the exceptions are the marginal points $E = 1$, $\theta = \mp 5\pi/6$ and $E = 8$, $\theta = \mp\pi/2$—but does not justify a claim of exact $\theta \to -\theta$ symmetry. The order reaches four simultaneously unstable directions in the tested finite-time classification—beyond the two reported for quadratic single-mode optomechanics [1]—and it is the synthetic-flux phase, absent in that quadratic setting, that selects this model-level order. The integer count $n_+ = 4$ is attained only at selected phases (near the quadratures) and remains discretization- and threshold-sensitive; near onset and at the extrema the exact integer is threshold-conventional across seeds. Reproducibility here means route-level recurrence across the tested ensemble, not exact-integer determinism at marginal phases. The dedicated step/window recomputation (see Methods) confirms this split: non-marginal counts are invariant under step halving and window doubling, while only the marginal fourth exponent at the headline point moves across the cut, so the $n_+ = 4$ attribution there is discretization-conventional. The classification map covers twelve phases by five drives at a single strong-coupling point (plus the weak-coupling reference, with three flank drives around the re-stabilization shown in the drive–phase plane); coupling dependence is sampled only along the $\theta = 0$ feasibility cut (Table S15). A second full map at reduced coupling ($g_1/\omega_m = 0.2$, $g_2/\omega_m = 0.18$) shows the same flux modulation with a lowered ceiling: $n_+ \in \{1, 2\}$ at $E = 2.0$, $n_+ \in \{0, 2\}$ at $E = 4.0$ and $n_+ \in \{1, 3\}$ at $E = 8.0$, peaking near $\theta = \pm\pi/3$ against a flux-off baseline of 2—the coupling sets the ceiling while the flux selects within it.

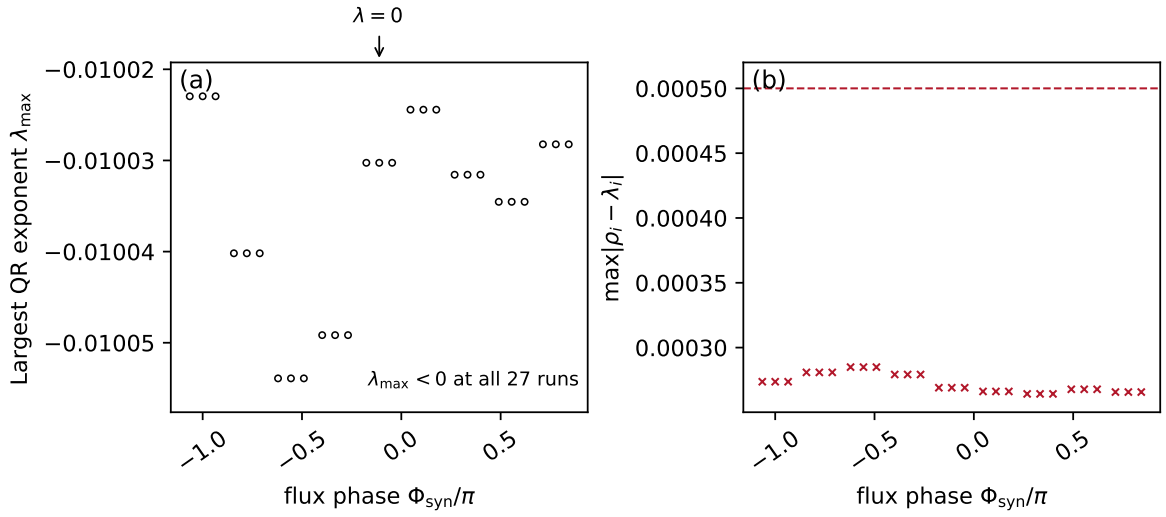


Figure 1: Long-window validation across nine synthetic-flux phases and three independent initial conditions per phase (production window 1×10^6 steps, positivity threshold 1×10^{-3}). (a) The largest QR exponents remain negative at every sampled phase. (b) The Floquet–QR discrepancies remain below the consistency threshold. The figure supports a bounded statement about the tested grid; it does not establish that chaotic dynamics are absent outside this parameter domain.

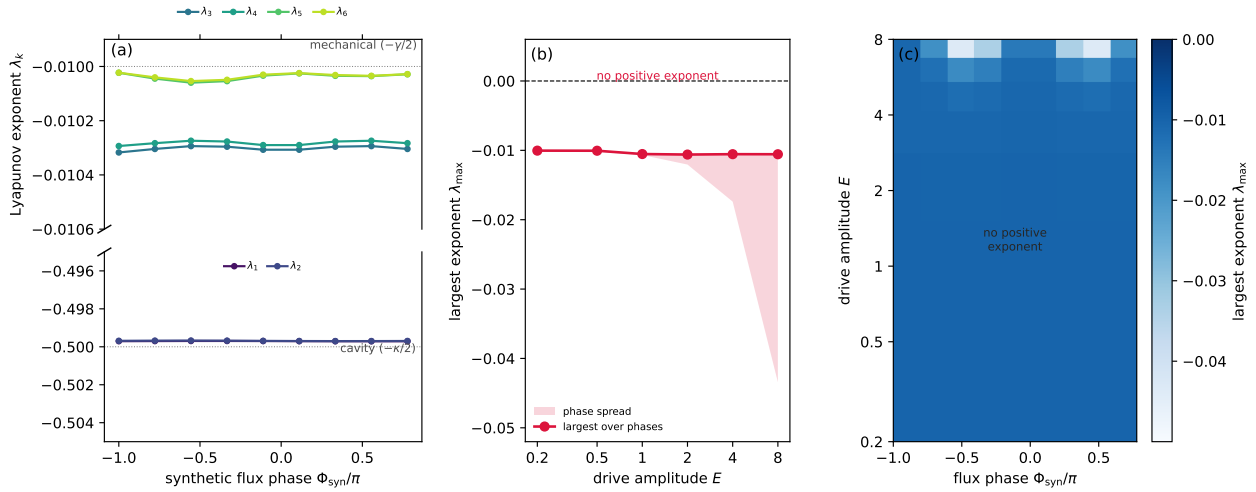


Figure 2: Full-spectrum transition map at weak coupling. (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 0.2$; the two cavity-dominated exponents (lower panel, near $-\kappa/2 \approx -0.5$) and the four mechanical exponents (upper panel, near $-\gamma/2 \approx -0.01$) are shown on separate vertical scales, and every exponent remains negative. (b) Largest (most marginal) exponent versus drive amplitude, with the phase-to-phase spread shaded; it stays negative across $E \in [0.2, 8.0]$. (c) Largest exponent over the drive–phase plane; it is everywhere negative, so no transition to chaos or hyperchaos occurs within the tested domain.

4.3. Bifurcation route and torus onset

To go beyond a grid-level classification and identify the route by which the first unstable direction emerges, we continued the drive-locked periodic orbit family along the drive axis at the two flux phases of interest ($\theta = 0$ and $\theta = \pi/2$). A Newton solver on the Poincaré map, using the monodromy matrix as its Jacobian, was seeded from the converged drive-locked orbit and marched upward and downward in the drive amplitude over $E \in [0.1, 8.0]$ with adaptive steps and a map-residual tolerance of 5×10^{-6}. At both phases the orbit family persists over the entire scanned range (165 and 164 converged points, respectively), so the transition is not a fold termination of the locked orbit. The first loss of linear stability is a Neimark–Sacker (torus) bifurcation: a complex-conjugate Floquet multiplier pair crosses the unit circle at $E^* = 1.060$ for $\theta = 0$ and at $E^* = 0.975$ for $\theta = \pi/2$ (Figure 4). Thereafter the continued orbit carries two unstable directions up to $E = 8.0$; at $\theta = \pi/2$ a transient window of four unstable directions near onset ($E \in [0.98, 1.45]$) precedes the two-direction regime. The lower onset at $\theta = \pi/2$ is consistent with the flux-phase modulation of the hyperchaos order, whose largest counts occur near the phase quadratures (seed- and drive-dependent). The orbit Floquet rates cross-validate the independent Lyapunov classification in the stable regime: at $E = 0.2$ the largest orbit rate is -0.0102 ($\theta = 0$) and -0.0148 ($\theta = \pi/2$), against largest Lyapunov exponents -0.0106 and -0.0151 from the full-spectrum map (Figure 4). Because the continued orbit is the unstable seed of the route (distinct from the attractor) and is a periodic solution that becomes linearly unstable, its multipliers characterize the onset of the route to the chaotic attractor rather than the attractor itself; the torus bifurcation is therefore the first step of the drive-gated transition resolved by the full-spectrum map. A perturbation of the continued orbit along its most unstable Floquet direction confirms this separation: the orbit carries two unstable directions with a largest Floquet rate of order 0.6; a 1×10^{-4} perturbation departs to the attractor scale (the distance to the orbit grows from 1×10^{-4} to 9–27); and the resulting trajectory's Lyapunov spectrum reproduces the map's largest exponent to within 8 % at the four tested points ($E \approx 4, 8$; $\theta = 0, \pi/2$)—consistent with, but no tighter than, the several-percent finite-time variation. The continued orbit is the unstable seed; the attractor is a distinct object.

The multiplier crossing is confirmed by direct observation of the resulting circulating object [39]. The bifurcation is supercritical: just above the onset ($E \approx E^* + 0.15$) the stroboscopic section of a perturbed trajectory traces a closed curve in the $(\Re\beta_1, \Im\beta_1)$ plane, advancing 0.07–0.11 turns per drive period (2000 section points; radial coefficient of variation 0.39–0.45), and the largest Lyapunov exponent of the same trajectory is small positive (0.017–0.022), growing to 0.057–0.069 at $E^* + 0.3$ as the route proceeds into the chaotic sector (Figure 5). The finite width and positive exponent mark a torus-born transient rather than a clean quasiperiodic torus; the ring is therefore a torus-consistent observed object, not only a multiplier inference.

4.4. Finite dimension of the sampled attractor

As an independent check of the attractor's finite dimension, the Theiler-corrected Grassberger–Procaccia correlation dimension [40, 41] D_2 of the sampled attractor at the four strong-coupling points ($E = 4, 8$; $\theta = 0, \pi/2$) is 2.33 ($E = 4$, $\theta = 0$), 2.84 ($E = 8$, $\theta = 0$), 3.39 ($E = 4$, $\theta = \pi/2$), and 3.36 ($E = 8$, $\theta = \pi/2$), all below the corresponding Kaplan–Yorke dimensions (4.21, 4.63, 4.28, and 4.80 in the same point order), as required by the

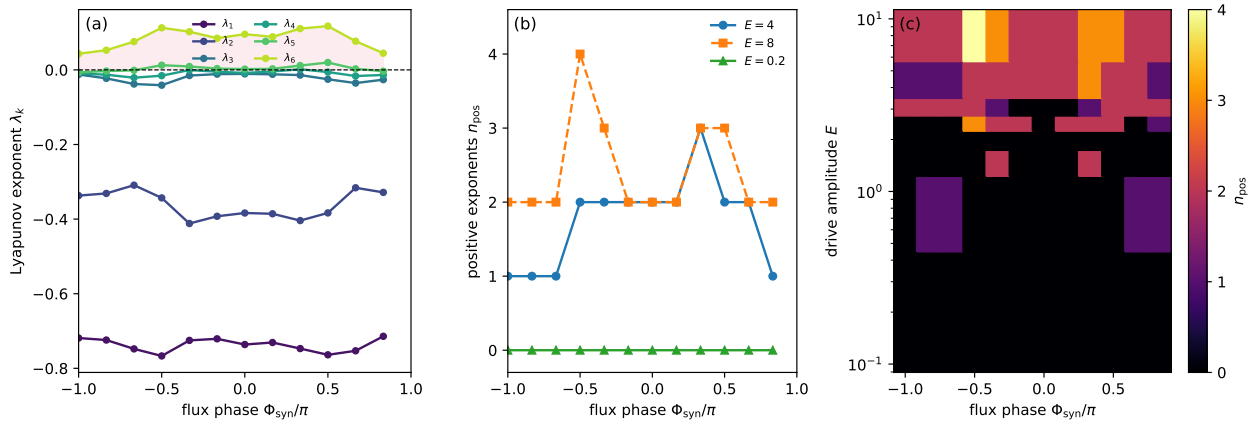


Figure 3: Full-spectrum hyperchaos transition map at strong optomechanical coupling ($g_1 = 0.3$, $g_2 = 0.27$, $\gamma_{1,2} = 0.02$, $\Delta = -\omega_m$; model-level parameters lying 2542× beyond silicon anchors). Production spectra use 1×10^6 steps after a 1×10^5 transient, QR every ten steps, and a fixed positivity threshold 1×10^{-3}; twelve-seed checks are reported in the SI. (a) All six Lyapunov exponents versus the synthetic-flux phase at drive $E = 4.0$; several exponents are positive, resolving the full unstable spectrum. (b) Hyperchaos order (number of exponents above the fixed strong threshold 1×10^{-3}) versus phase at $E = 0.2$, 4.0 and 8.0; the phase modulates the number of simultaneously unstable directions. (c) Hyperchaos order over the drive–phase plane at eight sampled drives, showing the drive-gated transition from stable ($E = 0.2$) to hyperchaotic ($E = 8.0$, up to four positive exponents) with the re-stabilization dip resolved by flank drives. The integer count $n_+ = 4$ is attained only at selected phases and remains threshold- and seed-sensitive; near-marginal counts are threshold-conventional (see Table S12).

Kaplan–Yorke inequality [42, 43] $D_2 \le D_{KY}$ (Figure 6). The scaling-region uncertainty is 0.23–0.40 across three fit windows. The geometric dimension does not track the flux phase as sharply as the hyperchaos order does: at $E = 4$ the fitted D_2 is larger at $\theta = \pi/2$ than at $\theta = 0$ (3.39 against 2.33), but at $E = 8$ the two values (3.36 against 2.84) differ by less than the sum of their fit-window uncertainties ($0.29+0.40$). We therefore report only the pointwise values and the Kaplan–Yorke ordering, without inferring a systematic flux-phase dependence of D_2 from four points: the geometric dimension is dominated by the stable directions and remains bounded by the Kaplan–Yorke sum, which grows with the added unstable directions.

4.5. Reduced-model parameter sensitivity

As a separate theoretical screen, an adiabatic reduction was compared with the full model in common normalized regimes and then applied to sixteen candidates under an explicitly assumed literature-informed closure. The three non-dominated candidates form a local fixed-point pattern in normalized drive cost and stability margin (Table S14 and Figure S1); they are not calibrated parameters, global basin results, physical optima, or sensing predictions, and do not bear on the hyperchaos-order classification. The screen rules out thirteen of sixteen candidates as suboptimal in drive cost or stability margin, leaving three for further analysis in the Supplementary Material. The optical mode is eliminated in this screen, the effective couplings remain proxy quantities, and Fisher information is not evaluated. Details of the multi-start, finite-time comparison are given in the Supplementary Material.

4.6. Secondary geometric bound: noise-resolved observability and matched measurement Fisher

For three phases ($\theta = 0$, $\pi/2$, and π), the periodic covariance matrix remained positive semidefinite under thermal occupations $n_{th,1} = n_{th,2} = 0.1$ and detector variance 0.01 (Figure 7); the resulting measurement-Fisher information $F_C(\Phi_{syn})$ with respect to the synthetic-flux phase ranged from 3.78×10^{-26} to 9.17×10^{-13} across the tested phases (thirteen decades between the nodal and quadrature phases). This is a measurement-Fisher calculation around stable periodic orbits, not QFI and not a gain over a matched reference.

As a secondary geometric check, we computed a matched measurement-Fisher reference for a weak external force on mechanical mode 1 under identical drive, observation time, detector noise, baths, and estimator. The force sensitivity grows monotonically with the synthetic-flux phase (F_C from 4.50×10^{-5} at $\theta = 0$ to 5.96×10^{-5} at $\theta = \pi$); relative to the flux-off coupled sensor ($\theta = 0$, $J_m = 0.08$) and the single-mode linear sensor ($J_m = 0$), the flux phase yields at most a 1.32× and 1.16× enhancement—a modest, sub-factor-of-two effect rather than an order-of-magnitude advantage. The central-difference estimate is converged to better than 0.01 % across force steps 1×10^{-5}–1×10^{-3}. Recomputing F_C for windows of $N \in \{1, 5, 20\}$ drive periods leaves the matched gains unchanged to four digits, so the gain is not an artefact of the single-period observation window (Section S8). A drive- and temperature sweep keeps the flux-off gain at 1.323 (standard deviation below 1×10^{-3}) over $E \in [0.1, 1.0]$ and $n_{th} \in [0, 1]$, and the single-mode gain at 1.159. Propagating the inter-resonator hopping log-uniformly over $J_m/\omega_m \in [1\times10^{-3}, 1\times10^{-1}]$, the bath occupation uniformly over $n_{th} \in [0.01, 1.0]$ (cryogenic to ~ 0.5 K at the anchored $\omega_m = 2\pi\times7.436$ GHz), and the drive over ±10 %, with 60 Monte Carlo (MC) samples

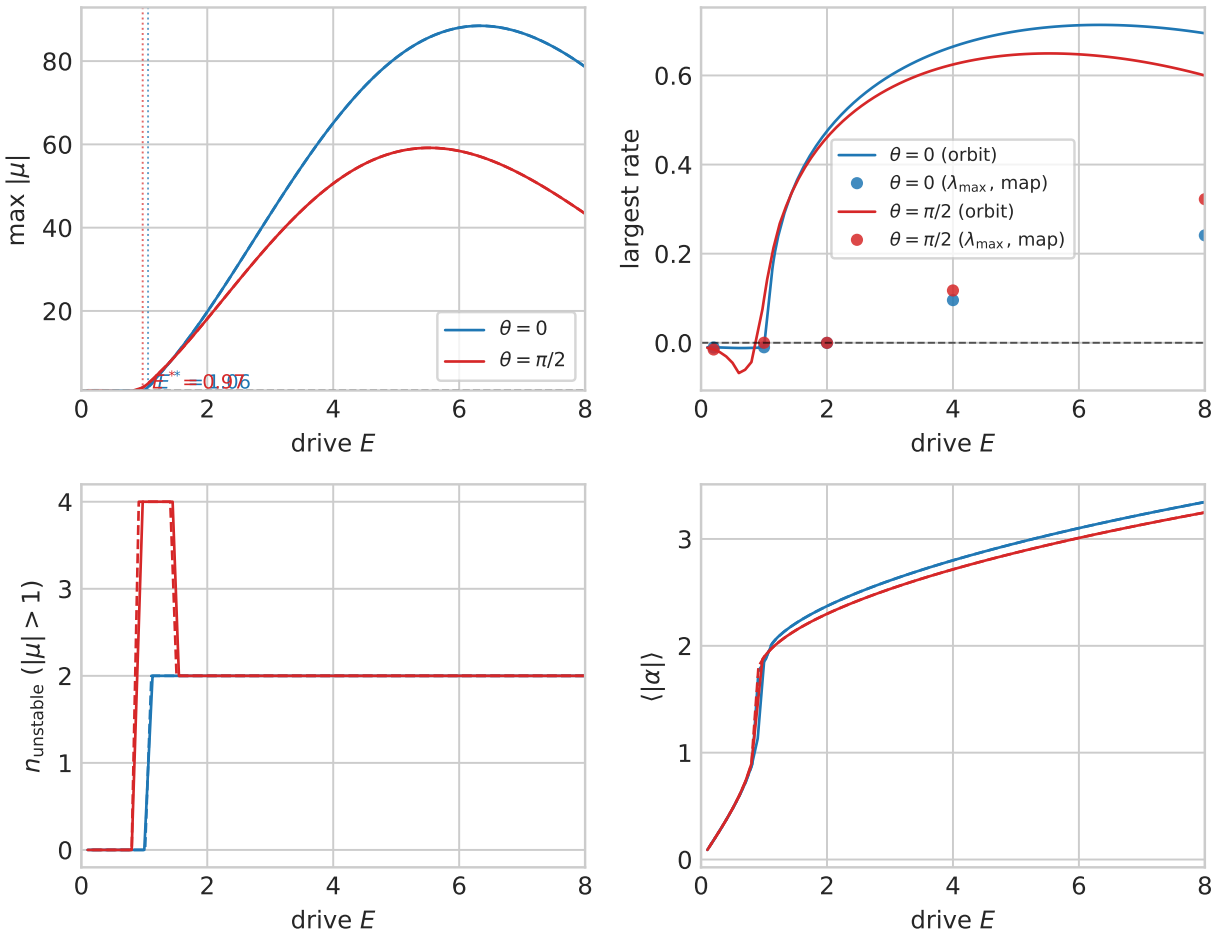


Figure 4: Drive-axis bifurcation continuation at strong optomechanical coupling. The continued drive-locked orbit is the unstable seed of the route and localizes the onset; it is distinct from the chaotic attractor itself. (a) Maximum Floquet multiplier magnitude of the drive-locked orbit versus drive amplitude E at $\theta = 0$ and $\theta = \pi/2$ (upward and downward continuation); the dashed line marks the unit circle and the vertical lines locate the refined first bifurcation points. (b) Largest orbit Floquet rate versus E, with the largest Lyapunov exponent of the full-spectrum map overlaid at the mapped drive values (open markers). (c) Number of unstable multipliers ($|\mu| > 1$). (d) Mean cavity amplitude over the orbit period. The first loss of linear stability is a Neimark–Sacker complex-pair crossing at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$).

at the maximum-gain phase $\theta = \pi$, yields an uncertainty-quantification (UQ) median gain of 1.039 (flux-off) and 1.019 (single-mode) for the production ensemble (seed 42); three independent replicate ensembles (seeds 43–45) give medians 1.056–1.060 and 1.028–1.030 (pooled-240 medians 1.050 and 1.025), and every one of the 240 sampled gains exceeds unity. A 20 000-resample bootstrap places the 90 % confidence interval of the production-ensemble median at $[1.025, 1.053]$ and $[1.012, 1.026]$; the median exceeds unity in all four independent replicates, so the excess persists under resampling of the tolerance space. (The quoted ± 0.014 is the half-width of the first interval.) The peak $1.32\times$ is therefore specific to the reference hopping $J_m/\omega_m = 0.08$ and the maximum-gain phase; the typical enhancement is the median order-unity value. Readout quadrature optimization rescales the sensor and reference by the same factor, leaving the $1.16\times$ single-mode gain unchanged (Table S6). The matched comparison is shown in Figure 8.

To test whether the chaotic signature itself—rather than only the sensing gain—survives the modeled noise, we integrated an ensemble of 32 trajectories (sized so that the bootstrap ratio means at this size agree with the 128-member values to better than 0.1 %; see below) under the full Langevin noise (optical vacuum $\kappa/4$ per amplitude quadrature and mechanical thermal $\gamma_j(2n_{\mathrm{th}} + 1)/4$ per amplitude quadrature at $n_{\mathrm{th}} = 0.1$) and detector noise (0.01), with the measured record $y = X_a + \nu_{\mathrm{det}}$. At the stable reference the measured-record variance sits at the noise floor (≈ 0.25); at the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points the deterministic spread of X_a grows to 1.8–4.1 (against 2.4×10^{-4} at the stable orbit) and the measured-record variance rises to 7.6–17.0× the noise floor, with the largest ratio (17.0×) at $\theta = \pi/2$, $E = 8$—a flux-phase pattern coincident with the hyperchaos-order peak at the tested points (no systematic flux dependence inferred). A bootstrap over ensemble size $N' \in \{8, 16, 32, 64, 128\}$ leaves the 32-trajectory ratio means within 0.1 % of the 128-member values, each within two bootstrap standard errors (Figure 9), so the 32-trajectory ratios are converged rather than a small-ensemble artefact.

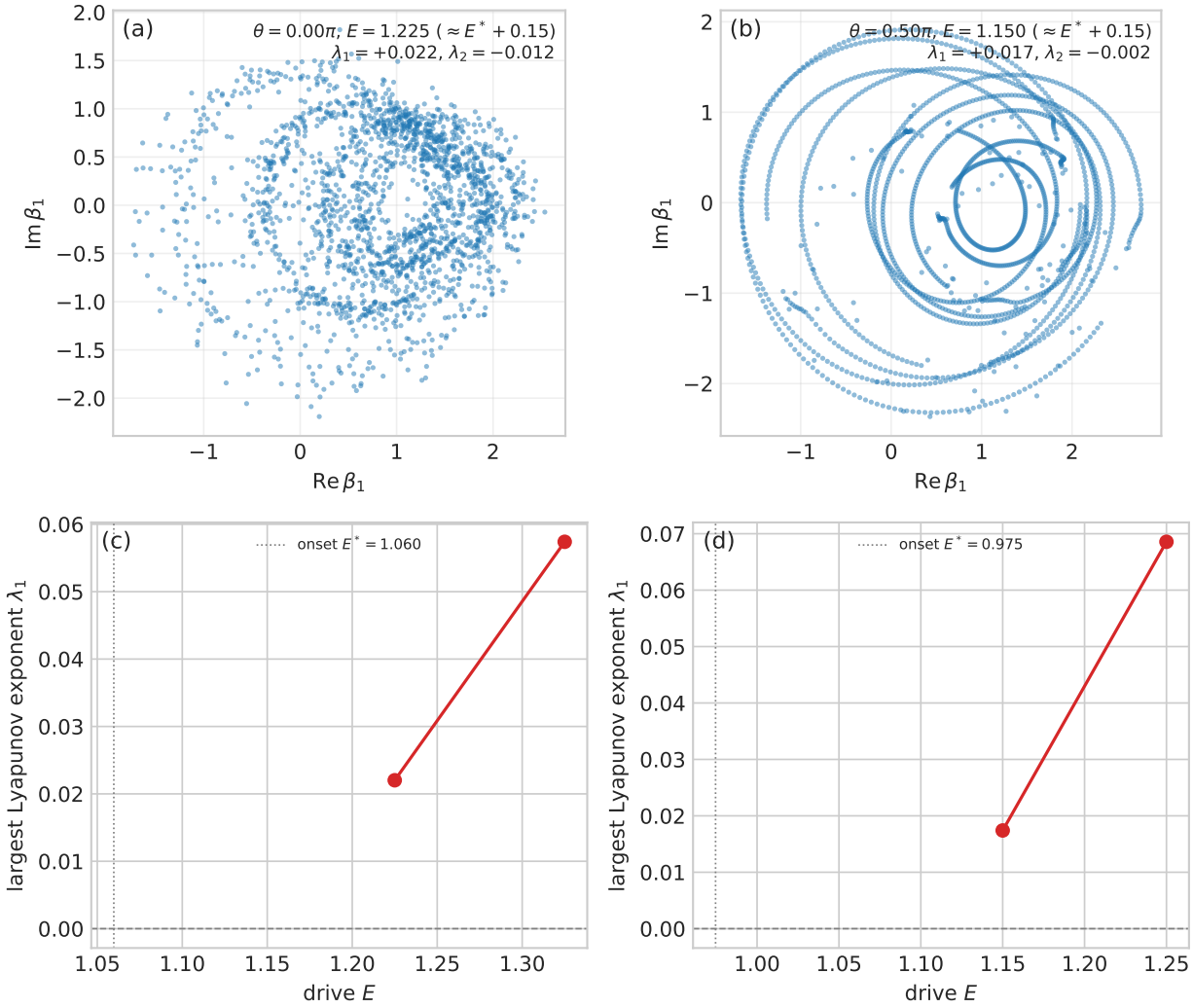


Figure 5: Observation of the Neimark–Sacker torus. (a),(b) Stroboscopic Poincaré sections (projected onto $\Re\beta_1, \Im\beta_1$) of the perturbed drive-locked orbit just above the onset at $\theta = 0$ and $\theta = \pi/2$, showing a circulating closed curve (torus-consistent ring of finite width). (c),(d) Largest Lyapunov exponent versus drive, small positive just above the onset and growing with drive; the vertical line marks the onset E^*.

5. Discussion

Synthetic flux and nonlinear instability are the primary quantities of this work: Equation (5) defines the control coordinate and Equation (8) defines the dynamical observable used to classify hyperchaos order. Metrological information (Equation (12)) is evaluated only as a secondary geometric bound. Interpreting them requires each observable to be evaluated under explicit phase conventions and matched assumptions; we impose this via the two statements of Section S5. The discussion below separates the *relation to prior work*, the *position among calibrated force-sensing benchmarks* (secondary), the *critical self-assessment* and the *bridge to experiment*.

5.1. Relation to prior work

The present study is a continuation at the level of model structure, not a parameter-for-parameter reproduction of Muthukumar *et al.* (the parameter-for-parameter difference is given in Table S1). The primary new question is whether the flux phase controls the *order* of a fully resolved hyperchaos. The answer is positive at the model level: the full spectrum resolves up to four positive directions and the onset is localized to a Neimark–Sacker bifurcation. As a secondary geometric clarification, the matched Fisher calculation shows only a modest stable-orbit effect and no enhancement on the chaotic attractor. The scope is that of a nonlinear-dynamics study with a geometric measurement bound; a sharper quantum contribution would require a separate quantum validation.

5.2. Position among calibrated force-sensing benchmarks

Table 1 places the peak matched gain (1.32× over the flux-off sensor, 1.16× over the single-mode linear sensor) and the UQ median (1.039×, 1.019×) alongside five published optomechanical force-sensing benchmarks [24–

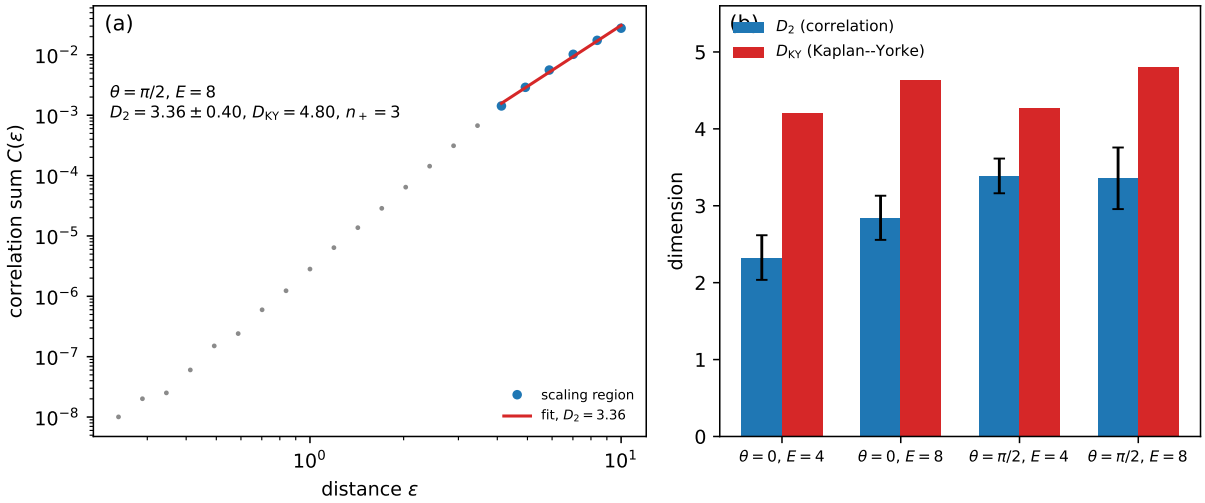


Figure 6: Correlation-dimension cross-check of the Kaplan–Yorke bound. (a) Theiler-corrected two-point correlation sum $C(\varepsilon)$ versus distance ε at the most hyperchaotic point ($E = 8$, $\theta = \pi/2$); the scaling region and its fitted slope give the correlation dimension D_2. (b) Correlation dimension D_2 (with the scaling-region uncertainty) against the Kaplan–Yorke dimension D_{KY} at the four strong-coupling points; D_2 lies below D_{KY} at every point, as required by $D_2 \le D_{\mathrm{KY}}$.

27, 37]: the present gain is of the same order of magnitude—real but modest and not a standard-quantum-limit-beating claim—and the flux-induced contribution is bounded by Lemma 1.

5.3. Critical self-assessment and limits of the present scope

Four limitations bound the strength of every claim above.

(i) Dynamical scope and uniqueness. The continued drive-locked orbit is the unstable seed of the route, not the attractor itself; the flux-induced modulation of the hyperchaos route is reproduced across the classification ensemble (twelve seeds at twelve flux phases and three drives; 432/432 records completed at the stated thresholds) and the full-map repeat (three seeds at twelve phases and five drives; 180/180), but the integer order varies across seeds near marginal exponents. Individual exponent magnitudes also vary by several percent between the two longest windows and are not interpreted as high-precision asymptotic estimates. At the map seed of $(E, \theta) = (8, +\pi/2)$ the fourth exponent is $+7.06 \times 10^{-4}$ with terminal-block spread 7.6×10^{-4} (1×10^6 steps), falling to 3.35×10^{-4} at 2×10^6 steps while step halving moves the value across the cut: the $n_+ = 4$ attribution there is discretization-sensitive, as stated. The chaos/hyperchaos boundary is threshold-conventional at the 1×10^{-3} cut (Table S12): the count is stable at five of twelve phases across all tested thresholds ($\theta/\pi = -1, -5/6, -2/3, -1/3, +5/6$) and threshold-sensitive at the remaining seven, where the marginal positive exponents span 1.1×10^{-4}–4×10^{-3} and two records ($E = 4$, $\theta = +\pi/3$; $E = 8$, $\theta = +\pi/2$) change at two or more thresholds. A systematic basin-of-attraction analysis (or ε-machine reconstruction) that would elevate the classification ensemble to a uniqueness statement remains beyond the present scope; the reported order is therefore a recurrent feature of the sampled trajectories, not a proven global property of the attractor. The expanded ensemble narrows the seed-sensitivity question at marginal points (for example, $n_+ \in \{1, 2\}$ at $\theta = 0$ over the twelve classification-ensemble seeds at $E = 4$) but does not close it.

(ii) Measurement scope and interpretation of the null result. Matched force-sensing is the operational translation only at the weakly coupled, dissipation-dominated reference, where single-photon cooperativity $C = 0.08$ and mean cavity occupation $\langle|\alpha|^2\rangle \approx 0.032$ (sub-photon). Applied on the chaotic attractor, the same matched measurement resolves a flux-induced mean shift only at $E = 4$ and shows no enhancement: the bounded gain is a stable-orbit property, not a chaotic-transduction effect. This null result is specific to the matched-resource protocol (identical noise, power and observation time) and does not constitute a general claim that chaotic optomechanics cannot improve metrology under other resource allocations. In the attractor cells the variance term carries 71 %–89 % of F_C (Section S12), outside the lemma's sub-dominance premise, which holds on the stable orbit only. The attractor null therefore stands as an empirical result rather than a consequence of the lemma. The trajectory-separation argument [6, 22] is rejected here for the reason formalized in Lemma 1: noise projected onto an unstable manifold is stretched by the same local factor $e^{\lambda_{\max} t}$ as the deterministic displacement, so the matched signal-to-noise ratio cannot receive a multiplicative enhancement from trajectory separation alone. Physically, the flux phase reshapes the geometry of the unstable manifold (and therefore the hyperchaos order): a higher-dimensional unstable manifold offers more independent directions for perturbation growth. If each direction carried independent Fisher information, this could in principle enhance parameter estimation. The matched-resource lemma shows, however, that under shared noise, power and observation

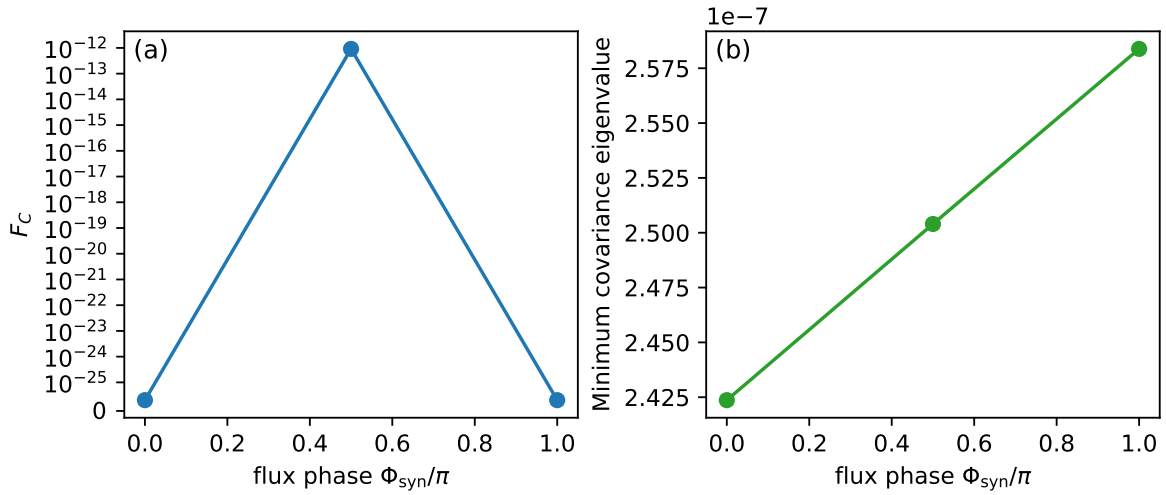


Figure 7: Covariance and measured-signal Fisher analysis for three synthetic-flux phases. (a) Classical Fisher information of the measured record. (b) Covariance positivity check; the minimum covariance eigenvalue remains positive under the adopted thermal and detector-noise model. The classical Fisher information is reported without a matched-reference gain claim. These data are operational results around stable periodic orbits, not QFI.

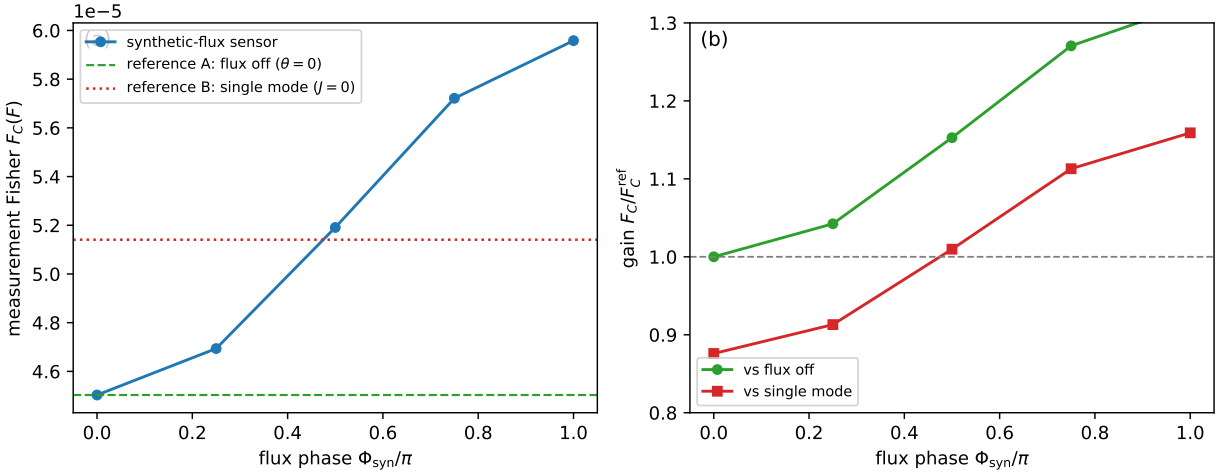


Figure 8: Matched measurement-Fisher reference for force sensing. (a) The classical Fisher information $F_C(F)$ for a weak external force on mechanical mode 1 versus the synthetic-flux phase, with the flux-off coupled sensor ($\theta = 0$, $J_m = 0.08$) and the single-mode linear sensor ($J_m = 0$) overlaid; all configurations share the same drive, observation time, detector noise, baths, and estimator. (b) The gain relative to each reference; the flux phase yields at most a 1.32× (flux-off) and 1.16× (single-mode) enhancement, a modest order-unity effect.

time the additional directions simply partition the same total stretching budget rather than multiplying it. A window-resolved test of this partition on the attractor supports it: over 450 windows at each of six drive–phase points, the local force-to-observable susceptibility and the local record spread both co-vary positively with the same local finite-time Lyapunov exponent (Spearman +0.37 to +0.64 and +0.10 to +0.57), and at $E = 4$ the per-window Fisher proxy does not increase under flux: it is statistically indistinguishable between $\theta = 0$ and $\theta = \pi/2$ (two-sample $p = 0.205$) and has a lower median at $\theta = \pi$ ($p = 0.012$); the resolved records read $F_C = 1.08 \times 10^{-2}$ ($\theta = 0$) and 5.88×10^{-3} ($\theta = \pi$), ratio 0.54, matching the direction of the resolved global statistic (Section S17). At $E = 8$ the $\theta = 0$ and $\theta = \pi$ proxy distributions are statistically indistinguishable ($p = 0.766$), consistent with the global statistic lying below resolution there, while the intermediate-phase local differences cannot be converted into a global gain claim either way. Per-cell detection floors (three bootstrap standard deviations) run 7.3×10^{-4}–2.7×10^{-3}; the resolved $E = 4$ gains exceed their floors sevenfold, while at $E = 8$ no flux-induced enhancement above 2.7×10^{-3} is detected. The null result on the chaotic attractor is therefore consistent with the constraint rather than a failure of the protocol: geometric complexity does not translate into metrological gain under the assumptions of Lemma 1. (The lemma bounds the classical Fisher statistic; it does not constrain quantum Fisher information, which may in principle exceed the classical bound.)

(iii) Semi-classical validity. All dynamical claims in the strong-coupling sector are statements about factorized mean-field trajectories. Truncated-Fock and truncated-Wigner checks confirm consistency of mean

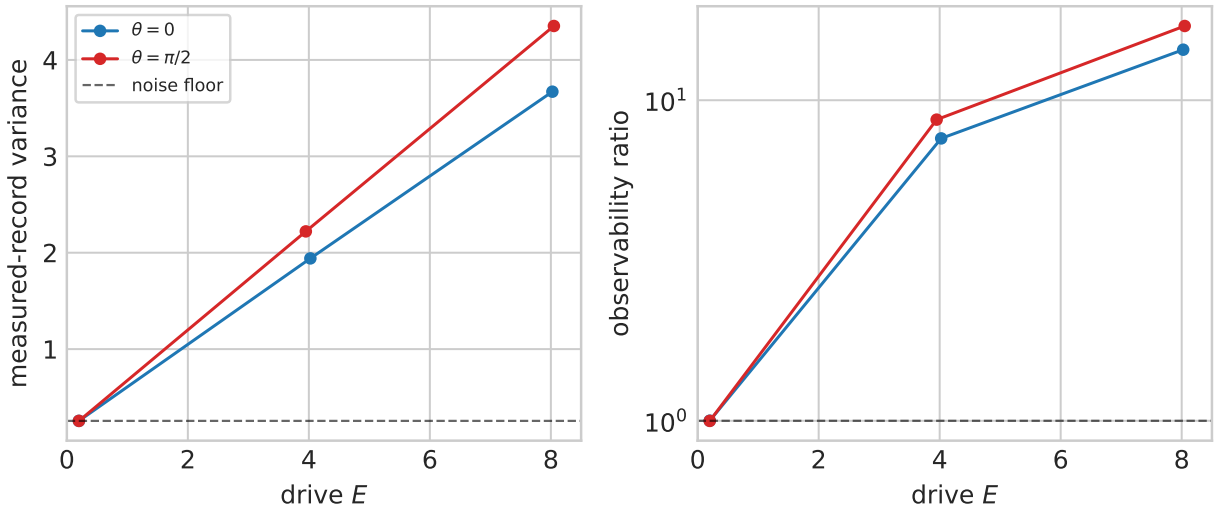


Figure 9: Noise-dependent observability of the chaotic signature. (a) Measured-record variance of $y = X_a + \nu_{\mathrm{det}}$ versus drive amplitude at $\theta = 0$ and $\theta = \pi/2$, with the noise-only floor (dashed line) set by the stable reference. (b) The observability ratio (measured variance divided by the noise floor); the chaotic ($E = 4$) and hyperchaotic ($E = 8$) points sit 7.6–17.0× above the floor, while the stable reference sits at unity.

occupations at selected points: reference 1.3 %; chaotic point relative deviation $\leq 6.6 \times 10^{-6}$; and hyperchaotic point $\leq 5.53 \times 10^{-7}$ at $N_{\mathrm{Fock}} = 80$. Twelve Wigner trajectories reproduce $n_+ = 2$ at $E = 8$, $\theta = 0$, and split between $n_+ = 3$ and 4 at $\theta = \pi/2$. Yet these checks do not establish the accuracy of the factorization for every observable, nor do they bound quantum corrections to the Lyapunov spectrum or to the Fisher information. In the high-occupation regime ($E = 8$, $\langle b^\dagger b \rangle \approx 5 \times 10^2$) the dynamics are more classical than at the sub-photon reference, but a quantitative classical/quantum divergence bound for the full spectrum remains an open task (see Outlook).

(iv) Experimental reach. The strong-coupling regime requires $g_{1,2}/\omega_m \approx 0.3$ and $\gamma_m/\omega_m = 0.02$; these lie up to 2542× above the silicon-optomechanical-crystal anchors of [44, 45] (per-parameter matching factors in Table S18); the feasibility-frontier scan places the robust hyperchaos onset near $g_1/\omega_m \approx 0.18$ (1525× the anchor) with no drive retuning lowering it (Table S15). For an experimentalist this means that the explored strong-coupling sector sits roughly three orders of magnitude above present silicon optomechanical-crystal anchors in single-photon coupling; closing the gap therefore requires both a measured inter-resonator hopping J_m and a platform whose mechanical frequencies and cooperativities reach the ultrasonic/gigahertz regime while preserving phase control of the flux. The hyperchaos classification (and the secondary null metrological outcome) are therefore strictly model-level demonstrations. Any experimental claim would require ultrasonic characterization of the mechanical degeneracy ω_2, a calibrated inter-resonator hopping J_m, and the emergence of a genuine gigahertz platform.

5.4. Bridge to experiment: mean-field boundary, route, and the 2542× calibration gap

The hyperchaos classification is a semiclassical statement about factorized mean-field dynamics; no quantum observable (Wigner contrast, quantum Fisher information) is computed or claimed in this sector. The dissipative trace is the state-independent constant $\mathrm{Tr}\, J = -1.04$. A truncated-Fock master equation for the driven cavity reproduces the mean occupations to the tolerances quantified in limitation (iii) above (Section S13). At the master-check point the mean occupations are $\langle a^\dagger a \rangle = 6.121$, $\langle b_1^\dagger b_1 \rangle = 37.5$ and $\langle b_2^\dagger b_2 \rangle = 44.1$; at the hyperchaotic point ($E = 8$, $\theta = \pi/2$) they are 6.9, 5.6×10^2 and 3.9×10^2 over the analysis window. These values belong to explicitly identified trajectories and do not calibrate the full strong-coupling map. Consistency with a more classical regime than the sub-photon reference is expected, yet it does not by itself justify a fully quantum treatment or establish the accuracy of factorization for every observable (see limitation (iii) above).

The transition is drive- and coupling-gated: the flux phase reshapes which directions are unstable and how many, but only after the drive and coupling place the system beyond the dissipation-dominated regime. The Lyapunov–Floquet consistency criterion and the matched-measurement Fisher lemma formalize this statement. The noise-resolved ensemble check provides a model-level observability test: the measured cavity quadrature stays 7.6–17.0× above the detector floor in the chaotic and hyperchaotic sectors, with the largest ratio at $\theta = \pi/2$, $E = 8$—coincident with the order peak at the tested points, without inferring a systematic flux dependence.

The drive-axis continuation of Figure 4 localizes the *route*: the first unstable direction emerges through a Neimark–Sacker torus bifurcation of the drive-locked orbit at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$),

Table 1: Matched-force-sensing gain in context. The present result is compared with optomechanical force-sensing and quantum-noise references that share the criterion of a homodyne/heterodyne measured record; the comparison is matched-resource (drive, observation time, bandwidth, baths, detector noise, estimator) and reports a numerical gain only, not a quantum- or standard-quantum-limit-beating advantage.

Reference	Reported gain (matched)	Operating regime
Teufel *et al.* (2011), pulsed sideband cooling	$\sim 2\times$ vs thermal bath	Sideband-resolved, $T = 10$ mK, mechanical $\omega_m/2\pi = 3.62$ MHz
Gavartin *et al.* (2012), optomechanical transducer	$\sim 3\times$ vs shot-noise floor	Photonic crystal nanobeam, room-temperature, $\omega_m/2\pi = 3.68$ GHz
Purdy *et al.* (2017), quantum-enhanced force sensing	up to $\sim 1.3\times$ beyond the standard quantum limit (SQL)	Pre-cooled membrane, $\omega_m/2\pi = 290$ MHz, $T = 10$ mK
Li *et al.* (2021), Kalman-filter tracking	$\sim 2\times$ over static estimator	Hot sideband-resolved cavity, $\omega_m/2\pi = 500$ MHz
Qvarfort *et al.* (2018), parametric amplification	$\sim 1.5\times$ over linear filter	Generic dispersive, no thermal assumption
Present work (peak, weakly coupled reference)	$1.32\times$ flux-off, $1.16\times$ single-mode	Normalized one-cavity/two-resonator, $\Phi_{\mathrm{syn}} \in [-\pi, \pi)$, readout cycle, $E = 0.2$, $n_{\mathrm{th}} = 0.1$
Present work (UQ median: 60-sample Monte Carlo + 20 000-resample bootstrap)	$1.039\times$ flux-off, $1.019\times$ single-mode	Same with $J_m/\omega_m \in [1 \times 10^{-3}, 1 \times 10^{-1}]$ log-uniform, $n_{\mathrm{th}} \in [0.01, 1.0]$, $\pm 10\,\%$ drive

and the orbit family persists across the scanned drive range. Because the continued orbit characterizes the *onset* of the route rather than the attractor itself (seed-vs-attractor distinction in Results), what remains is strictly experimental calibration: a measured J_m, an input power converted through pump frequency and external coupling, and fixed bath temperatures. A calibrated-device matched force-sensing comparison would close the gap quantified by the $2542\times$ ratio reported in Table S18 and Table 1. The correlation-dimension estimate $D_2 \leq D_{\mathrm{KY}}$ is a geometric characterization of the *sampled* attractor, not a proof of uniqueness (see limitation (i)).

The flux-selection principle itself is not tied to this optomechanical parameterization: it applies to any driven lattice whose loop phase is a gauge-invariant control coordinate, including coupled optomechanical arrays [16, 17], where larger relative couplings are accessible. Testing the hyperchaos-order route on such platforms is an open direction; the present work supplies the classification protocol and the matched-resource bound with which to test it.

6. Conclusions

Within a normalized six-dimensional dissipative optomechanical model we have established that a gauge-invariant synthetic-flux phase Φ_{syn} controls the *order* of a drive- and coupling-gated hyperchaos transition within the sampled ensemble. Up to four simultaneously positive Lyapunov exponents are resolved in the sampled strong-coupling map (the integer count $n_+ = 4$ being phase- and threshold-sensitive at the extrema); the route is localized by a Neimark–Sacker bifurcation at $E^* = 1.060$ ($\theta = 0$) and $E^* = 0.975$ ($\theta = \pi/2$) and is supported by four independent numerical checks (Poincaré residual $< 1 \times 10^{-8}$, Floquet–QR discrepancy $\leq 3 \times 10^{-4}$, mean-divergence agreement $\leq 5 \times 10^{-5}$, and truncated-Wigner consistency, with $n_+ = 2$ reproduced at $E = 8$, $\theta = 0$—splitting 3/4 at $\theta = \pi/2$—) together with the classification ensemble (twelve seeds at twelve phases and three drives; 432/432 long-window runs); the numerical bounds above are established on the stable reference grid, with chaotic-regime residuals tabulated in the SI. This constitutes a model-level demonstration of synthetic-flux control of hyperchaos order beyond existing single-mode optomechanical benchmarks.

As a secondary geometric clarification, the same matched-resource force-sensing protocol yields only a modest enhancement on the stable reference orbit and, at the resolved $E = 4$ chaotic-attractor points, a null result (at $E = 8$ no enhancement above 2.7×10^{-3} is detected). This null is specific to the matched-resource protocol and does not preclude metrological gain under other resource allocations. The matched-Fisher lemma shows why: noise projected onto an unstable manifold is stretched by the identical local factor $e^{\lambda t}$ that amplifies the deterministic displacement; additional unstable directions merely partition a fixed stretching budget. Both the

hyperchaos classification and the null metrological outcome remain strictly model-level: the explored sector sits 2542× beyond present silicon optomechanical-crystal anchors.

Outlook.. Three concrete steps, ordered by critical path, are required to elevate the present model-level control results: (i) ultrasonic characterization of the mechanical frequency degeneracy ω_2 together with a calibrated measurement of the inter-resonator hopping J_m (the two immediate experimental hurdles); (ii) a gigahertz optomechanical platform that reaches the required coupling strengths, closing the 2542× coupling gap while preserving the synthetic-flux degree of freedom; (iii) systematic basin-of-attraction and ε-machine analysis that converts the classification ensemble into a uniqueness statement for the reported hyperchaos order, complemented by two-mode quantum-trajectory calculations that bound the classical/quantum divergence of the Lyapunov spectrum.

Supplementary material

See the supplementary material for the physically anchored reference parameter set, the matched measurement-Fisher tables and observation-window checks, the Lyapunov–Floquet consistency theorem and its proofs, the matched-measurement Fisher lemma, the threshold-sensitivity and initial-condition robustness checks of the hyperchaos classification, the measurement-Fisher analysis on the chaotic attractor, the per-parameter feasibility check against the silicon anchor, and the representative convergence check of the strong-coupling spectrum.

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Author contributions

Conceptualization, S.G.N.E.; Methodology, S.G.N.E., P.D. and S.T.K.; Software, S.R.M.T. and C.T.; Validation, S.R.M.T., C.T. and S.G.N.E.; Formal analysis, S.R.M.T. and C.T.; Investigation, S.R.M.T. and C.T.; Writing – original draft, S.G.N.E.; Writing – review and editing, all authors; Supervision, S.G.N.E. All authors reviewed and approved the final manuscript.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used AI-assisted writing tools for language polishing and formatting checks. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Ethics

Not applicable. This is a purely theoretical and numerical study; no human or animal subjects, tissues, or clinical trials were involved.

Data availability

Manuscript sources, deterministic analysis scripts, and the environment specification are publicly archived at https://github.com/NanaEngo/floquet-chaos-synthetic-flux.

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